

# Welcome to Code Crunch!

## C12 | Crunch 3D: CAD & Modeling

[Tues 4-5PM] Weekly | Spring 2025

### Attendance

Earn a Certificate of  
Completion by attending  
at least 70% of the  
workshops



CODECRUNCH.CS.FIU



# 305 Hackathon : Edition Spring 2025



OPEN TO ALL STUDENTS, REGARDLESS OF MAJOR  
—NO EXPERIENCE NEEDED! FRESHMEN,  
GRADUATES, AND ALUMNI FROM ANY UNIVERSITY  
ARE WELCOME. LET'S INNOVATE TOGETHER!

Funded by SGA  
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# 305

SPRING 2025

# HACKATHON

HYBRID

**MAR 24 - 27 & MAR 29 - 31, 2025**  
WORKSHOPS | COMPETE | IN-PERSON CAREER FAIR

**FIU**  
Honors College

**hr** **epichire**

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#CODECRUNCH-HACKS

# Spring 2025 Weekly Schedule





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## CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Your CS Club for Collaboration, Networking, and Hands-On Projects!  
Open to all students, regardless of major—no experience needed! **Freshmen, graduates, and alumni from any university are welcome.** Let's innovate together!

|                                                                                         |                                                                              |                                                                         |                                                                              |                                                                           |                                                                                   |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>C5</b><br>Crunch AI<br>Data Science<br>Python<br>MON 1 - 2 PM<br>WED 12:30 - 1:30 PM | <b>C3</b><br>CodeCrunch<br>LABS<br>Portfolio<br>Java & React<br>MON 2 - 3 PM | <b>C7</b><br>Wire Crunch:<br>Embedded<br>Systems<br>C++<br>TUE 1 - 3 PM | <b>C11</b><br>Crunch<br>Arcade<br>Unity<br>TUE 3 - 4 PM<br>FRI 4 - 5 PM      | <b>C12</b><br>Crunch 3D:<br>CAD &<br>Modeling<br>OnShape<br>TUE 4 - 5 PM  | <b>C9</b><br>Swe<br>Convos<br>Networking<br>WED 1:30 - 2:30 PM<br>FRI 2 - 3PM *** |
| <b>C1</b><br>Crunch<br>Convos<br>Python<br>WED 2:30 - 3:30 PM                           | <b>C2</b><br>Crunch<br>The Code<br>Python<br>WED 3:30 - 4:30 PM              | <b>C6</b><br>Cybersecurity<br>Crunch<br>VM/LINUX<br>THU 1 - 2 PM        | <b>C4</b><br>CodeCrunch<br>LABS<br>Hackathon<br>Java & React<br>THU 2 - 3 PM | <b>C8</b><br>CodeCrunch<br>LABS:<br>Web Dev<br>Javascript<br>THU 3 - 4 PM | <b>C10</b><br>Crunching<br>Bioinformatics<br>Python<br>FRI 3 - 4 PM               |

**Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!**

RSVP on  
LU.MA/CODECRUNCH

In person or  
Stream on  
Zoom



 [CODECRUNCH.CS.FIU.EDU](https://codecrunch.cs.fiu.edu)

 **January 6, 2025 - April 4, 2025**  
Cost: No-cost

Become a member





# Unit #2

Topic: 3D Part Modeling

C12 | Crunch 3D: CAD & Modeling

# Key Dates and Structure

## C12 - Crunch 3D: CAD & Modeling

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- ★ **Week 1: Introduction to OnShape, Sketching Fundamentals, & 2D Constraints**

Learning what On-Shape is and how to properly sketch on OnShape.

- ★ **Week 2: 3D Part Modeling**

Learning how to use extrude, remove, fillet, chamfer, revolve, shell, and loft functions for 3D models.

- ★ **Week 3: 3D Part Modeling Continued**

Learning how to use draft, mirror, and sweep functions for 3D models.

- ★ **Week 4: Assemblies & Mating Constraints**

Learning how to create assemblies through mates and constraints.

- ★ **Week 5: Final Project - Real-World Design Challenge**

Applying all concepts to design a functional product.

- ★ **Week 6: Final Project Showcase**

Presenting your final project, sharing your experience, and receiving feedback.



# 3D Modeling Definitions

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- **Profile** is a closed 2D shape/sketch.
- **Faces** are the different sizes of a 3d Model.
- **Mirror** is used to reflect a sketch entity across a line/axis or a 3D model across a plane. It is useful for creating symmetric parts.
- **Extrude** is a feature that converts a profile into 3D shapes.
  - **Base Extrude** is the first extrude done in the model.
  - **Boss Extrude** is any extrude done after the Base Extrude.
  - **Remove** (a.k.a. Extrude Cut) uses a sketch to remove a section from an extrusion.



# 3D Modeling Definitions

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- **Fillet** is a modifier that rounds the edges or faces in a 2D sketch or 3D model.
- **Chamfer** is a modifier that created bevels on edges or faces in the model.
- **Revolve** is a type of extrude that creates a model around an axis.
- **Shell** is used to hollow out an extrusion. It leaves a consistent thickness for all the walls.
- **Loft** connects two or more profiles to form a 3D model or surface.

# Session Overview

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In this session we will be creating a link, a wheel, and a wine glass to practice the 3D modeling basics of extrusion, remove, fillet, chamfer, revolve, shell, and loft. For all models, we start off with a 2D sketch.

## Link:

Stick-like members that connect to each other. They can be used to make a movement (linear or rotational).

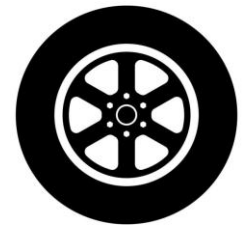
To create a link, we will be using extrude, remove, fillet, and chamfer.



## Wheel:

To create the tire for a wheel, we will be using revolve and shell.

This PowerPoint also demonstrates how to create a rim for the wheel using extrude.



## Wine Glass:

To create a wine glass, we will be using extrude, loft and shell.

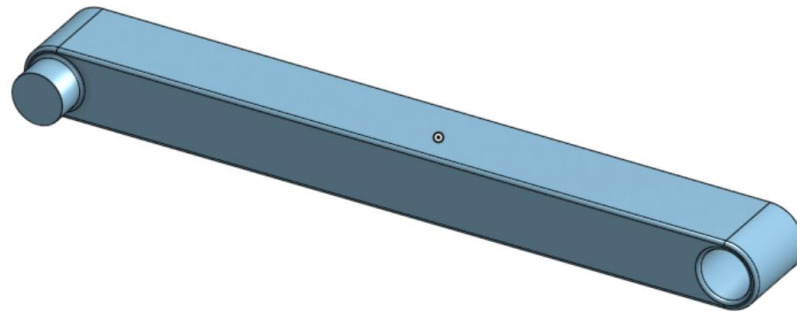




# Time for the...

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## Link!

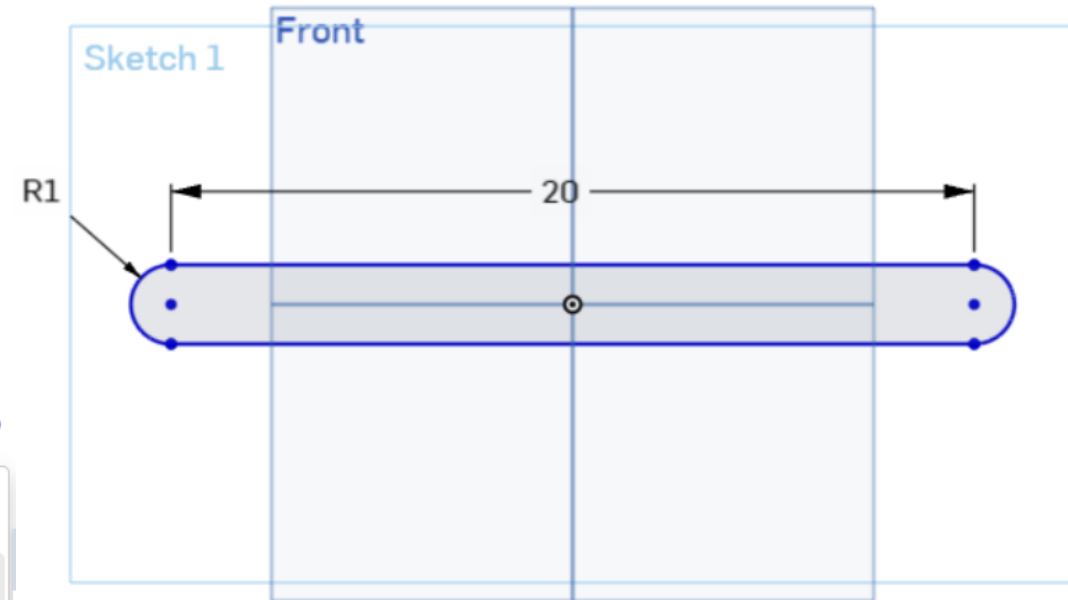
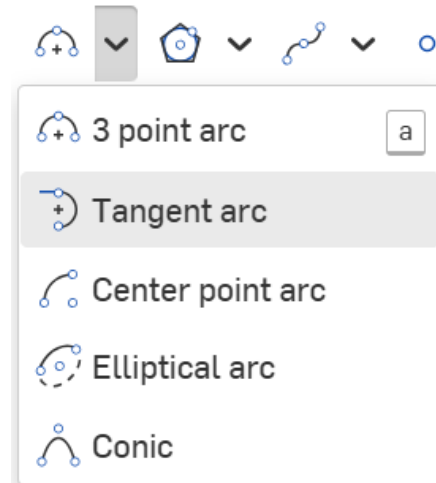


# Link: Sketch (& Arcs)

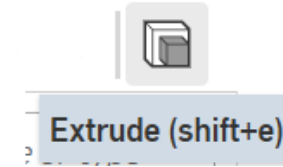
To start with the **link**, we are going to do a **sketch** on the **front plane**. The drawing consist of a **center rectangle** with the sides **trimmed**, and 2 **tangent arcs**. The width of the rectangle is 20 cm, and the radius of the arcs is 1 cm.

Note: Arcs allow you to make curves. A **tangent arc** allows for a smooth transition between the arc and another line/curve.

Types of arcs:

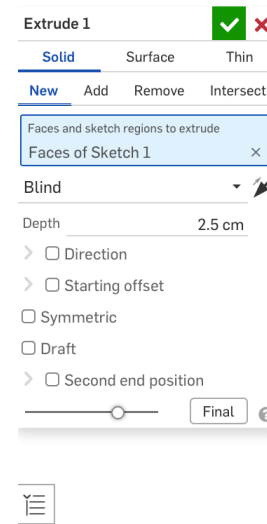


# Link: Extrude Base

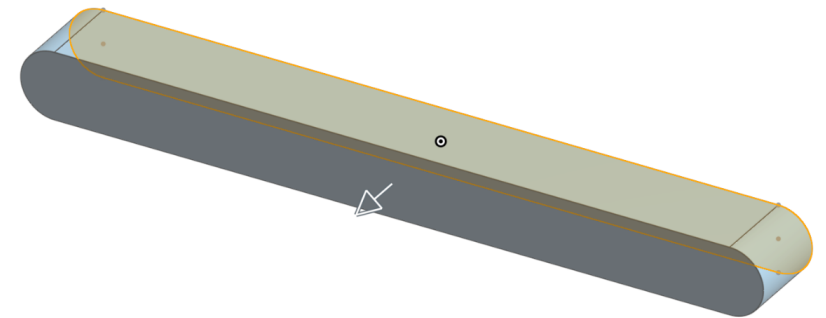


After the sketch is done, click the **green check mark** on the **sketch menu**.

Then, select the sketch on the left side and click **extrude** on the top. On **depth**, write 2.5 cm, and click the **green check mark** to make the 2D sketch a 3D model.

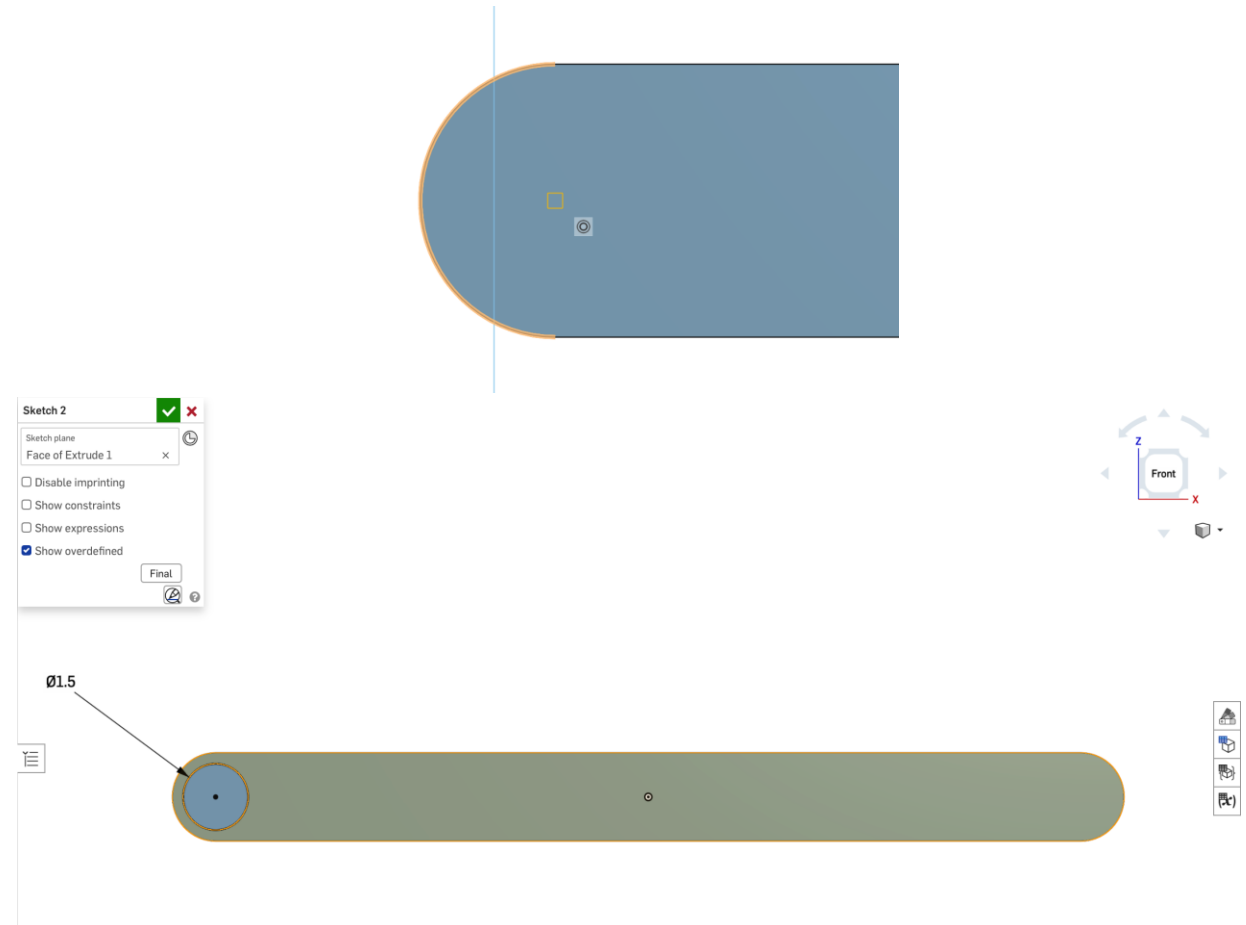


Note: when you finish the sketch, you can go directly to extrude by clicking the extrude feature at the top.



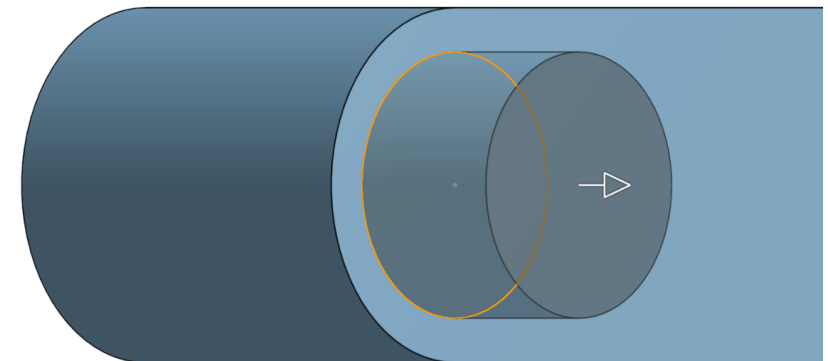
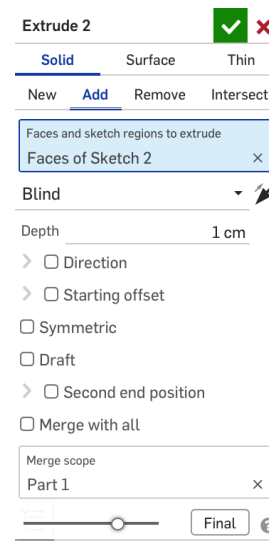
# Link: Sketch on a Body

Just like in a plane, you can do **sketches on faces**. Select the **face** (in this case, the front one) and do a sketch. Draw a **center point circle** in the **center of the arc** (indicated by the top point of the arc) with a radius of 1.5 cm.



# Link: Extrude Boss

Once the sketch is completed, click **extrude**, and change the **depth** to 1 cm, and click the **green check mark / Enter key**.

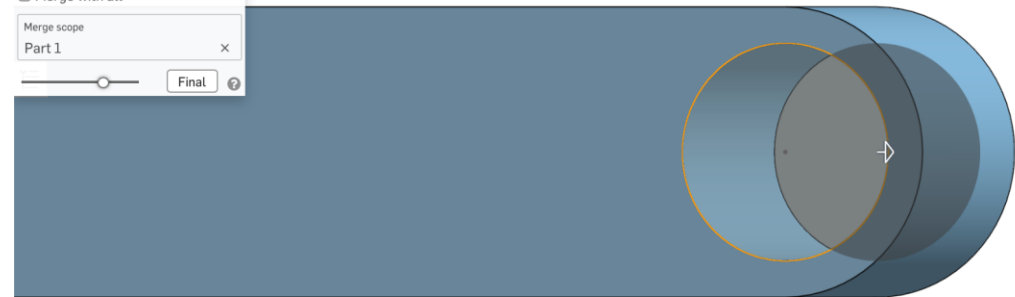
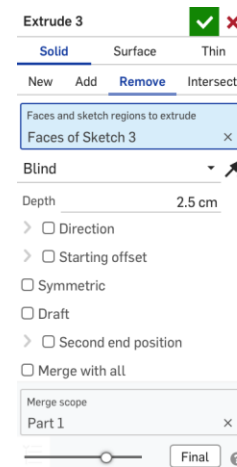




# Link: Remove

Now, we do the same circle sketch on the other side of the bar. After we have the sketch, click **extrude**, change from "add" to "remove", and click enter.

Note: The remove feature went through the entire link because 2.5 cm is the **default** measurement. If a hole is not desired and rather an **indent**, the depth must be changed.



# Link: Fillet

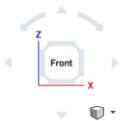
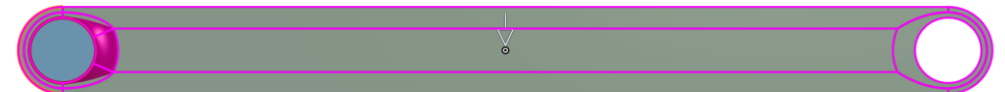
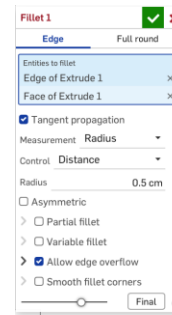
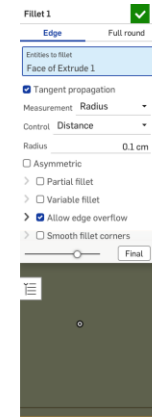


A **fillet** is used to rounds the edges or faces in 2D sketch or 3D model.

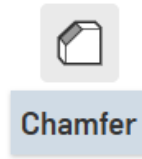
To do a **fillet** for the front face of the link:

1. Select the front **face** where you want to add it
2. Change the **radius** to 0.1 cm
3. Click the **green check mark**

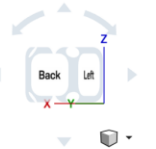
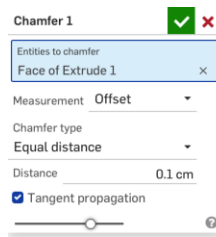
Note: When the **fillet is too big**, OnShape will indicate it with **pink lines**. If any changes are necessary, select the **feature**, right click it, and select **edit**.



# Link: Chamfer



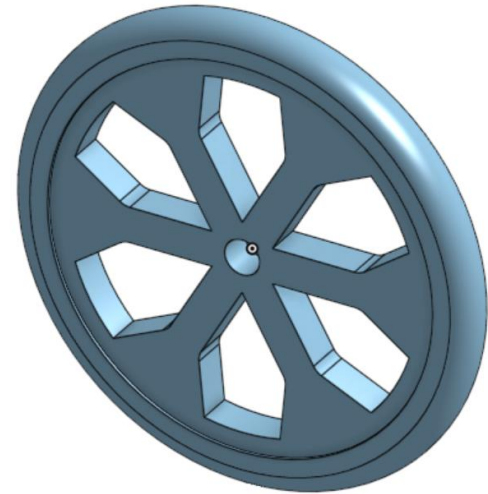
**Chamfer** is very similar to **fillet**.  
Select the back face of the bar  
and add a **chamfer** with a  
**distance** of 0.1 cm.



# Time for the...

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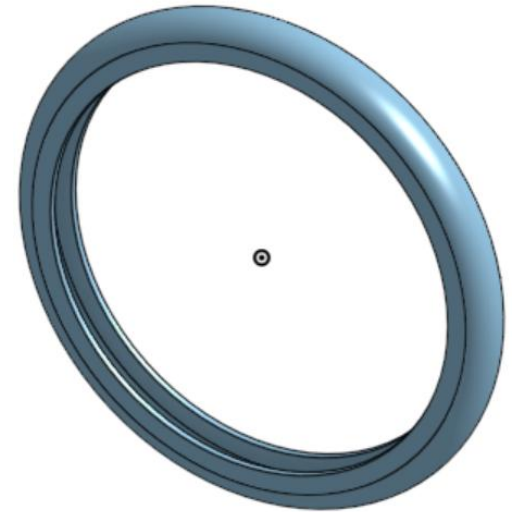
**Wheel!**



First...

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**Wheel Tire!**



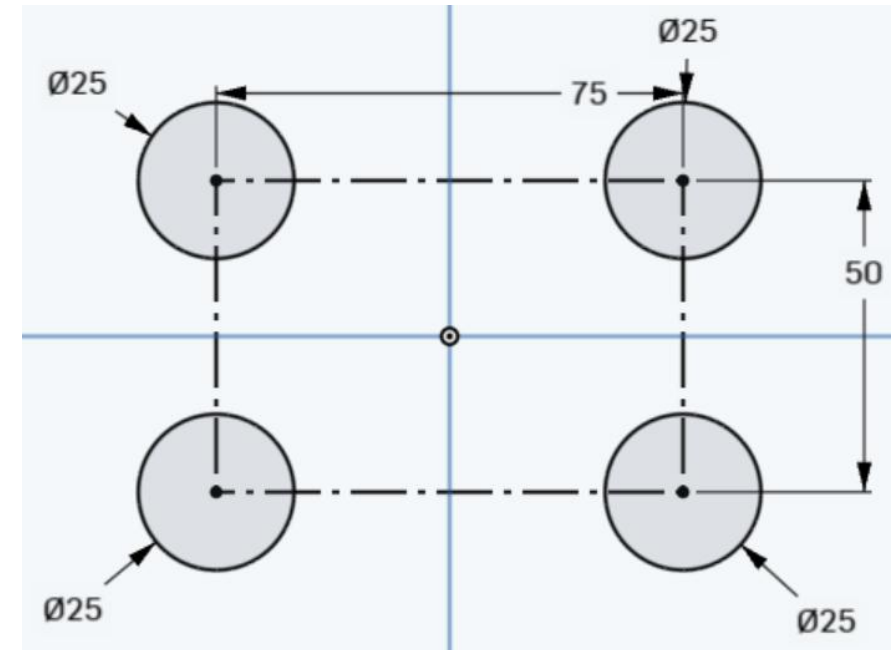
# Sketching: Construction



Sketch entities can be made into **construction** lines/curves.

**Construction** lines/curves are used as a **reference** or a **guide** for other components of a 2D sketch, 3D part, or assembly. They are **not** used to create features.

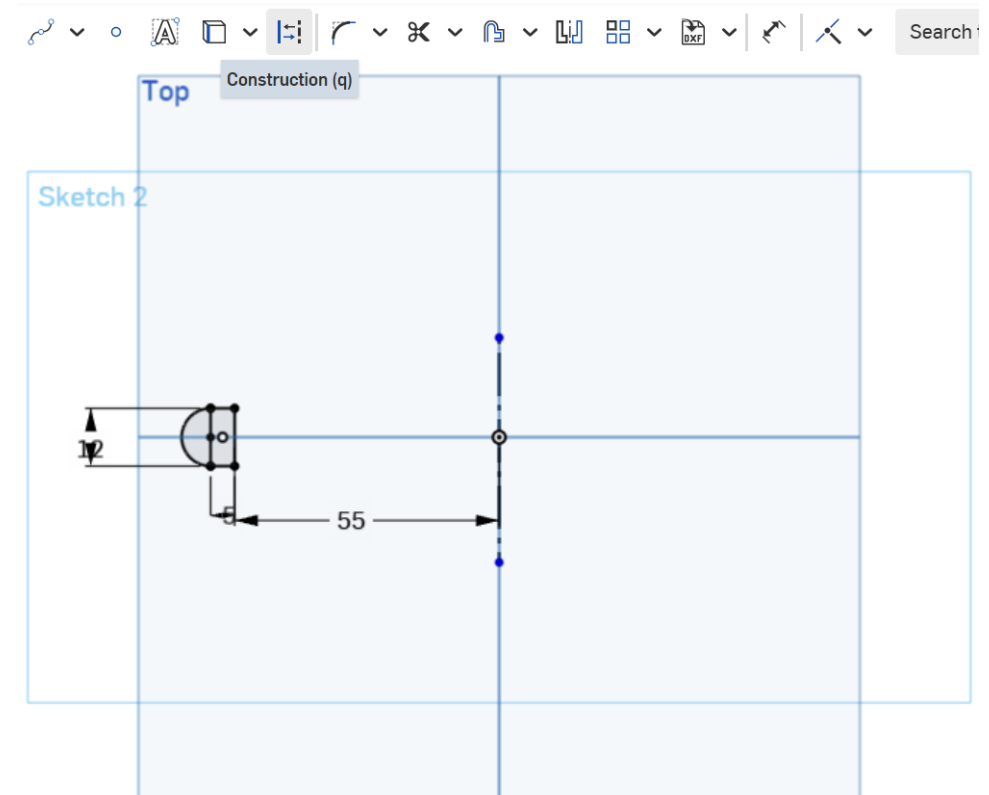
The image to the right displays a construction center point rectangle.



# Wheel Tire: Sketch

To start with the **wheel**, we are going to do a **sketch** on the **top plane**. The drawing consist of a **center rectangle**, a **tangent arc**, and a **construction line**.

The **center** of the **center point rectangle** is made to align with the horizontal axis, and it is **displaced** 55 mm to the left of the origin. A **tangent arc** is added to its left side. The rectangle has a **width** of 5 mm and a **height** of 12 mm (this automatically makes the tangent arc have a radius of 6 mm). Lastly, a **construction line** is drawn vertically through the origin (its length does not matter).





# Wheel Tire: Revolve



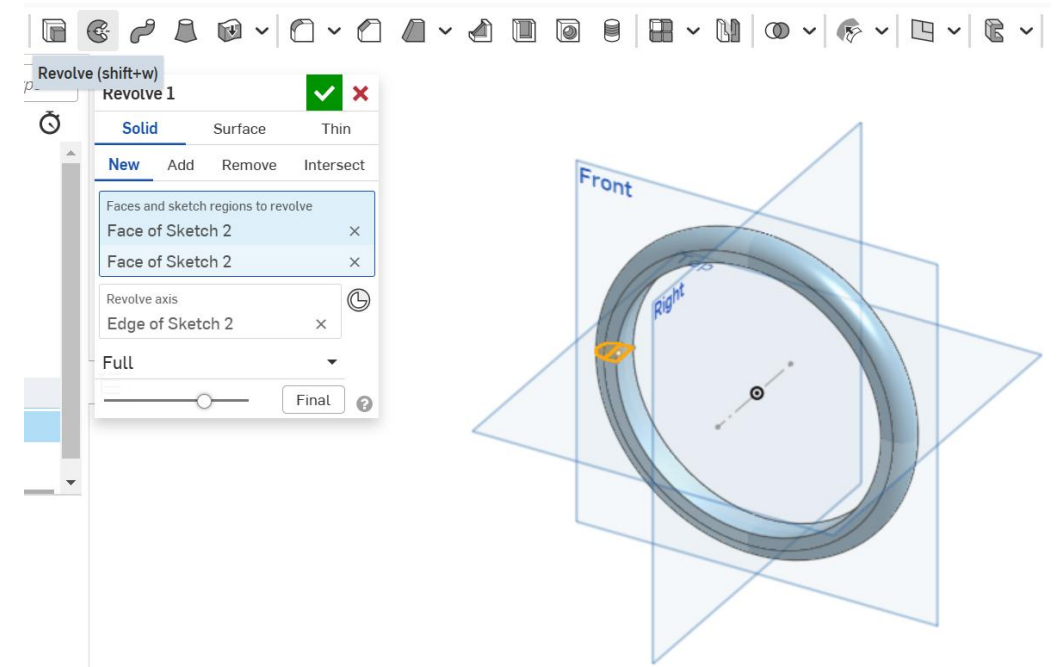
Revolve (shift+w)

**Revolve** allows you to rotate a 2D sketch about an axis of your choice.

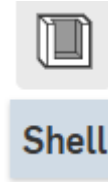
For the wheel, we revolve our 2D sketch about the construction line.

Notes:

- Ensure that “**solid**” is selected for the revolve to make sure it is treated as a 3D part.
- As demonstrated, the construction line was used as a guide for the revolve feature, but it was not used as a component of our 3D model.



# Wheel Tire: Shell

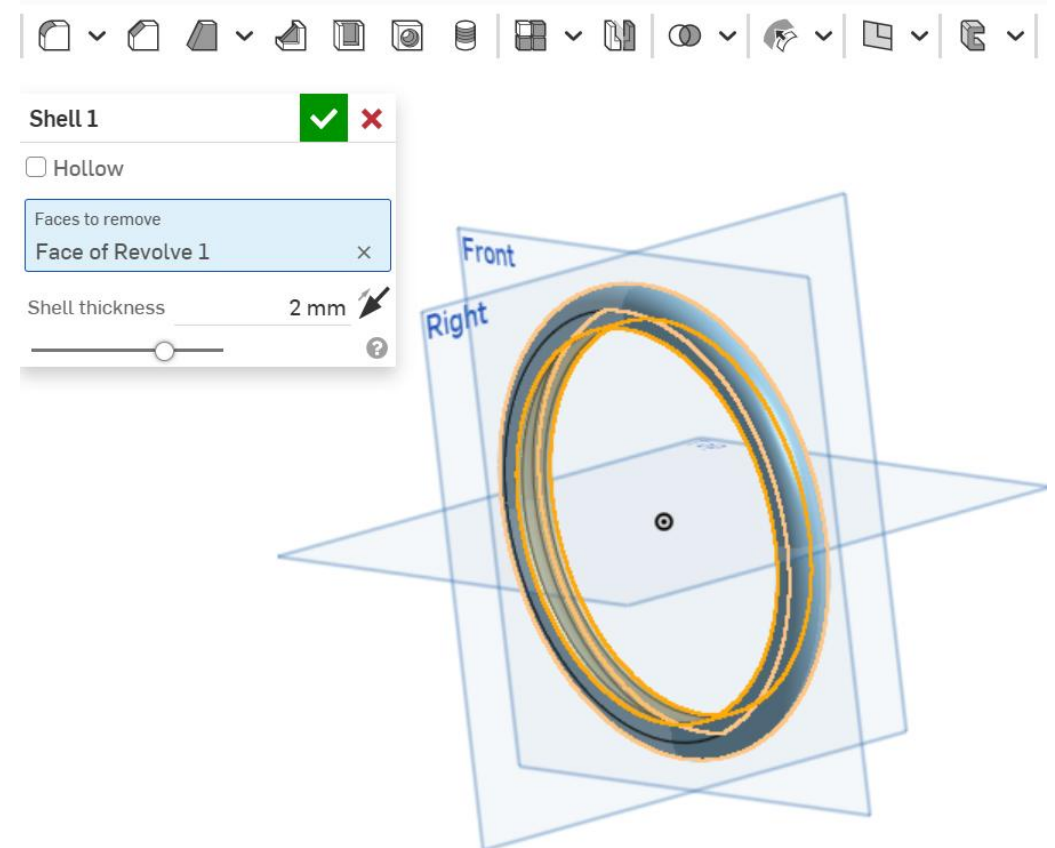


**Shell** is used to hollow out an extrusion with a consistent thickness for all the walls.

For the wheel, we used shell to hollow out the tire.

Steps:

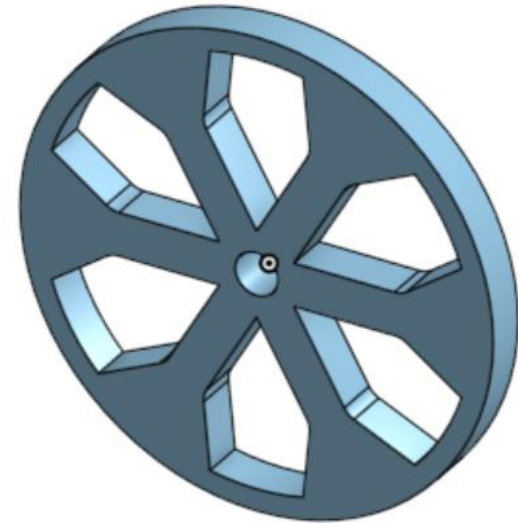
1. Click **shell**
2. Click the **inside** cylinder of the wheel
3. Make **shell thickness** 2mm
4. Click the check mark to complete your tire



# Extra: Then...

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## Wheel Rim!



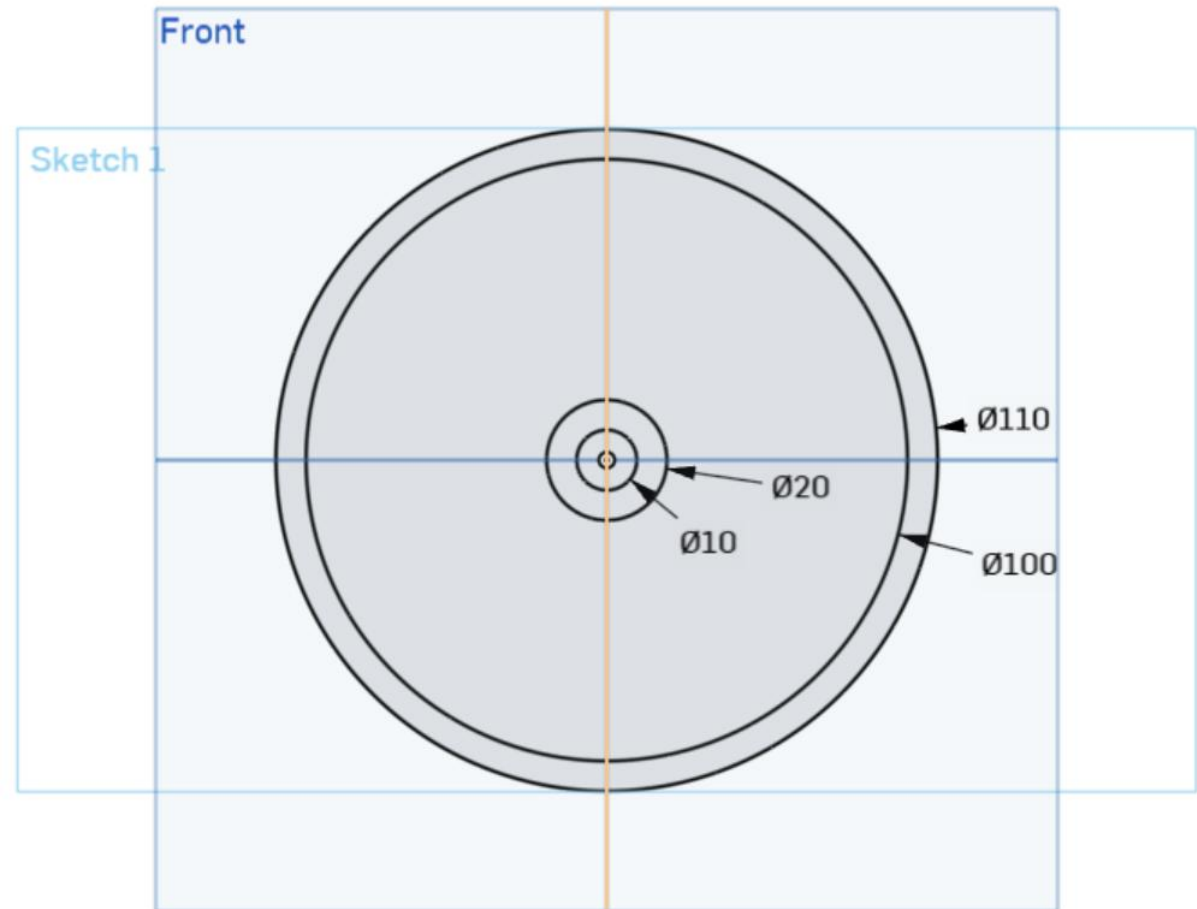
# Wheel Rim: Sketch Pt. 1

The **sketch** on the **front plane** is the main feature for the rim of the wheel.

Step 1:

Create 4 **center point circles** with their centers at the origin. **Diameters:**

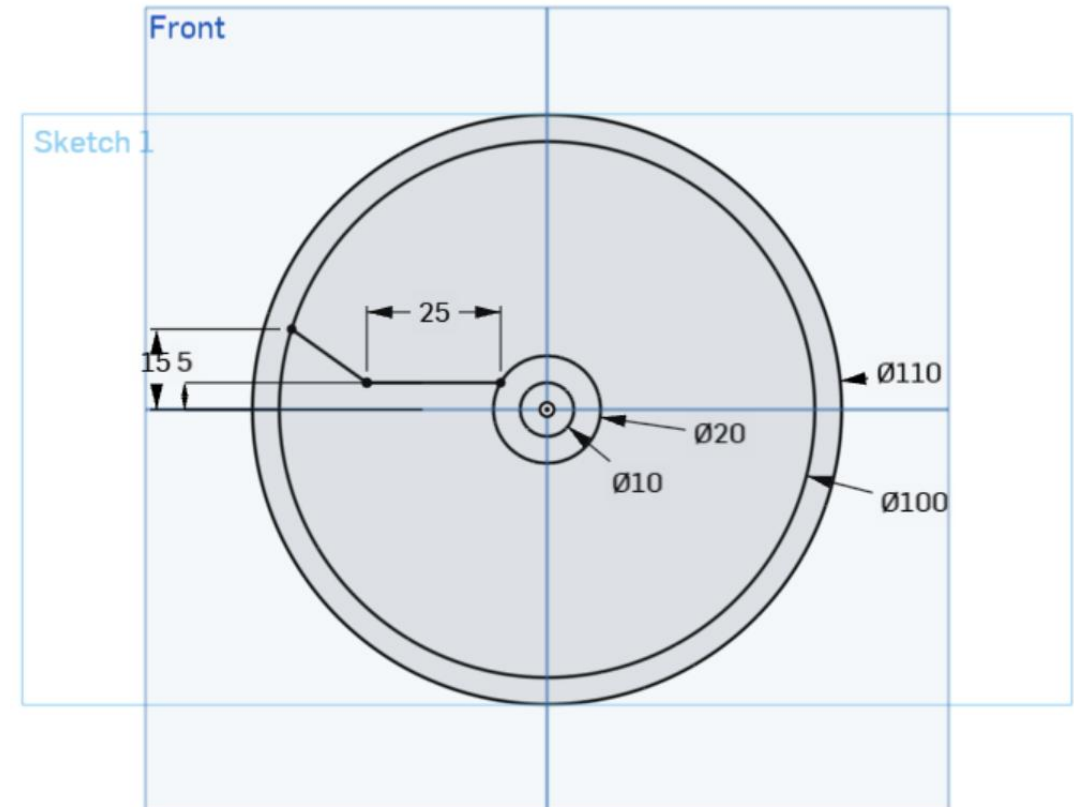
1. 110 mm
  - Note:  $2 \times 55 \text{ mm} = 110 \text{ mm}$ . This is why the inner radius of the wheel tire is 55 mm.
2. 100 mm
3. 20 mm
4. 10 mm



# Wheel Rim: Sketch Pt. 2

Step 2:

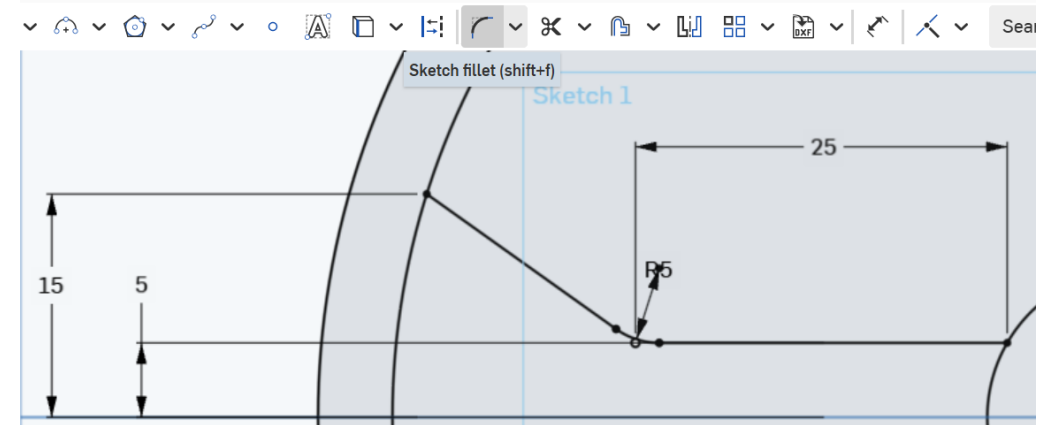
1. Create a **horizontal line** of length 25 mm that touches the 3<sup>rd</sup> circle (20 mm). Make it 5 mm away from the horizontal axis.
2. Make another **line** connecting the horizontal line to the 2<sup>nd</sup> circle (100 mm). Make the point at the end of this line that touches the 2<sup>nd</sup> circle be 15 mm away from the horizontal axis.



# Wheel Rim: Sketch Pt. 3

Step 3:

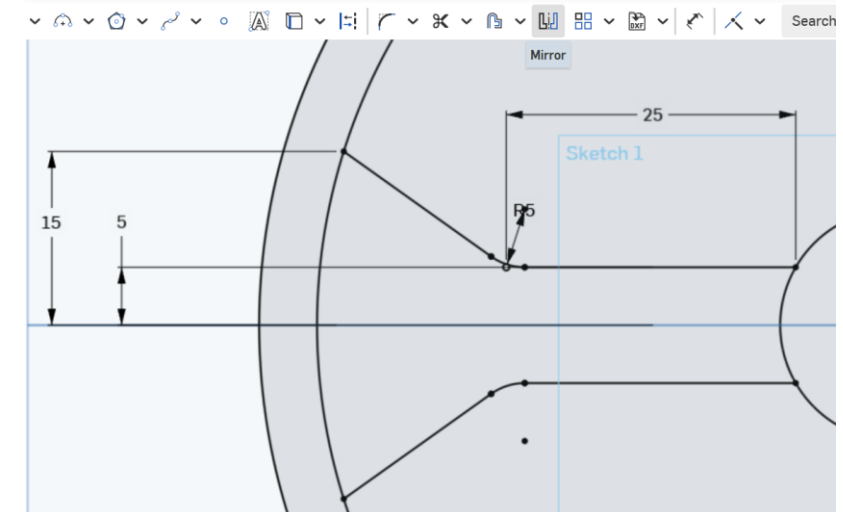
Make a **sketch fillet** with a **radius** of 5 mm at the point where the 2 lines intersect.



Step 4:

**Mirror** the 2 lines and fillet about the horizontal axis.

Note: **Mirror** is used to reflect sketch entities across a line/axis. In this case, select **mirror**, then the **mirror line** is the horizontal axis, and the 2 lines & fillet are the **entities to be mirrored**.



# Wheel Rim: Sketch Pt. 4

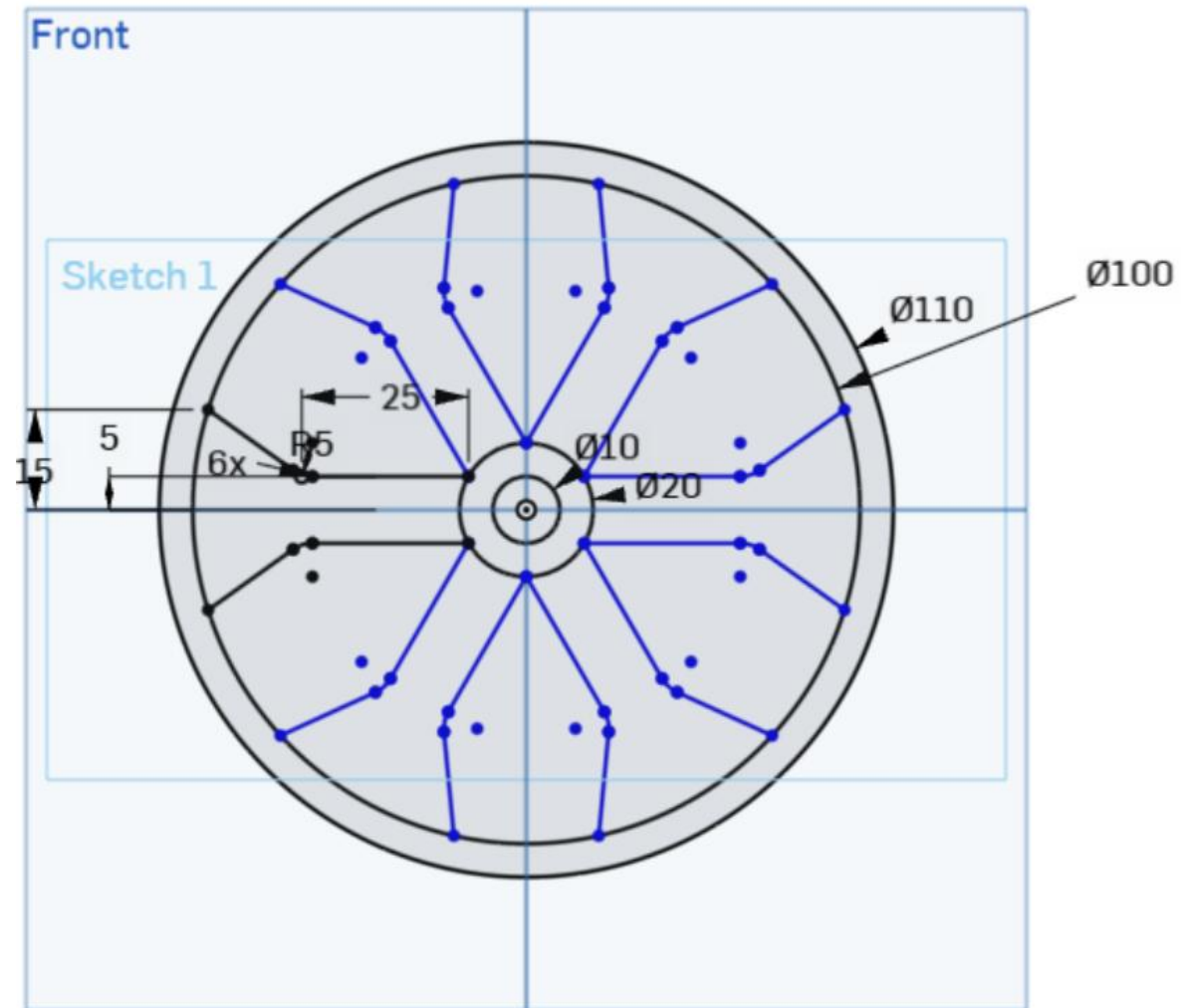
Step 5:

Create a **circular pattern** of the 4 lines and 2 fillets in between the circles.

Make them repeat 6 times.

Step 6:

Click the green check mark to complete the sketch.

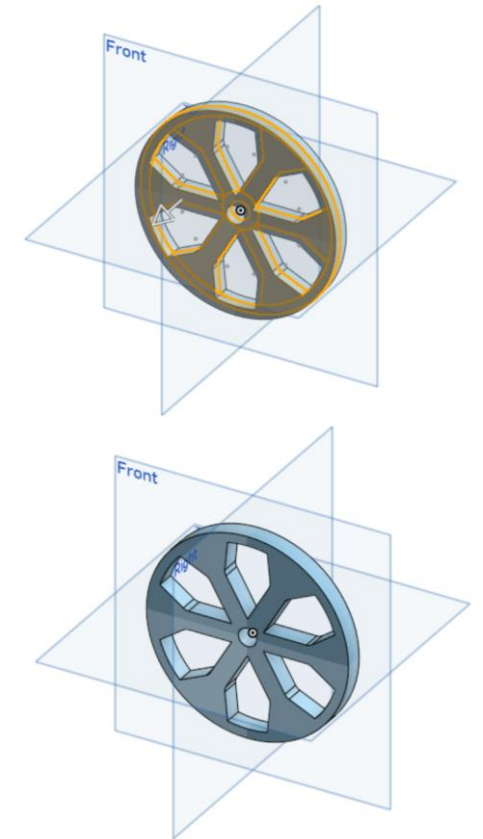
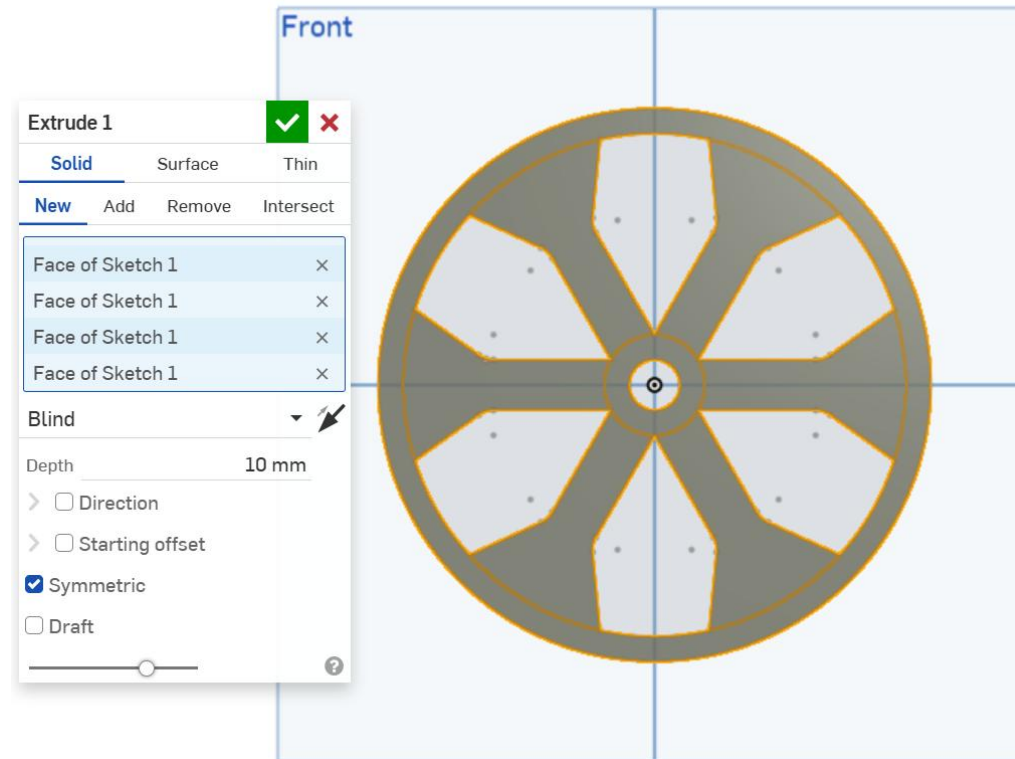




# Wheel Rim: Extrude (using symmetric)

To make the rim a **3D model**, we are going to **extrude** the sketch and make the **depth** 10 mm.

Check **symmetric** to make the rim extrusion have an equal **depth**/thickness on both sides of the **front plane**.

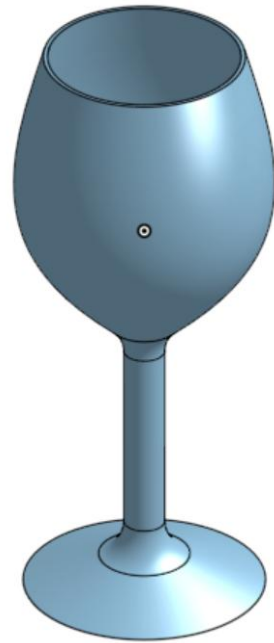




Time for the...

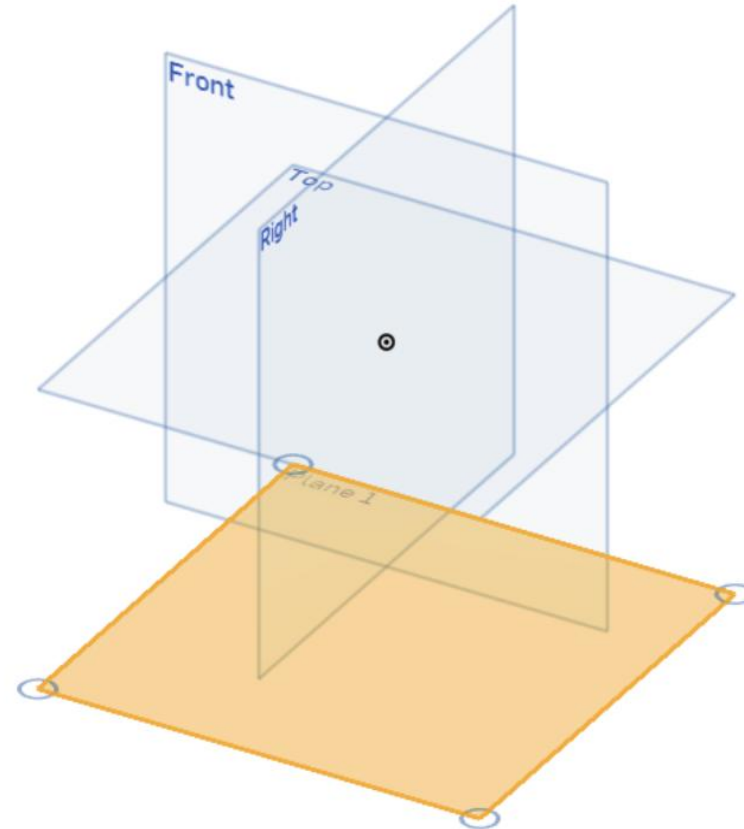
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**Wine Glass!**



# Offset Plane

Before we can start with the **Wine Glass**, we need to learn about **offset planes**. **Offset planes** are copies of the **original planes** at a **distance**. To create an **offset plane**, click “**plane**”,   select the plane you want to make a copy of (front, top, or right), and choose an **offset distance** and the **direction**.

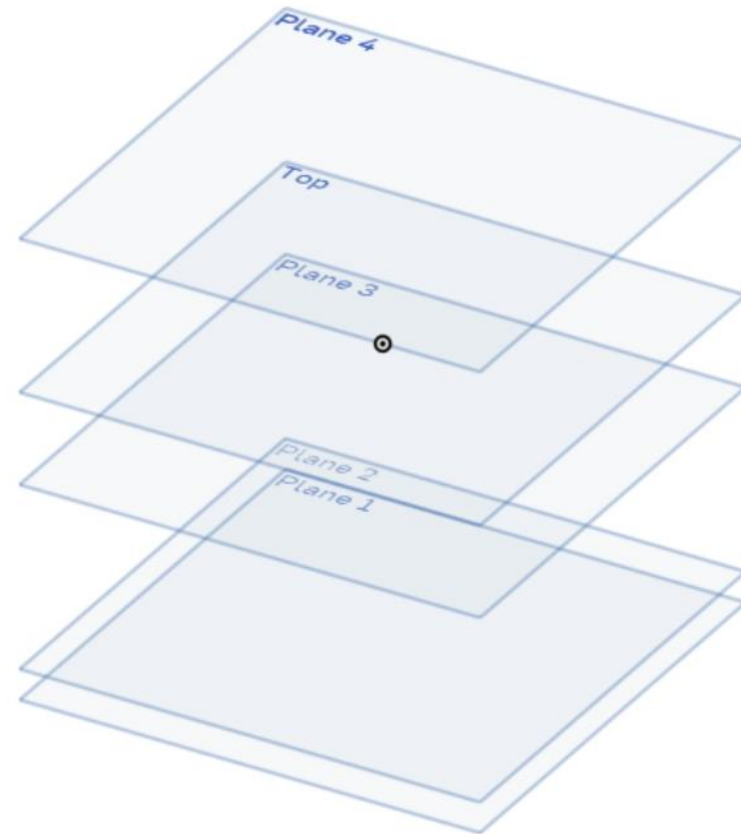


# Wine Glass: Offset Plane

For the **wine glass**, we are going to need 4 **offset planes** of the **top plane**.

1. Plane 1: 10 cm downwards
2. Plane 2: 9 cm downwards
3. Plane 3: 3 cm downwards
4. Plane 4: 5 cm upwards
  - Note: The front and right plane were hidden for a better visualization.

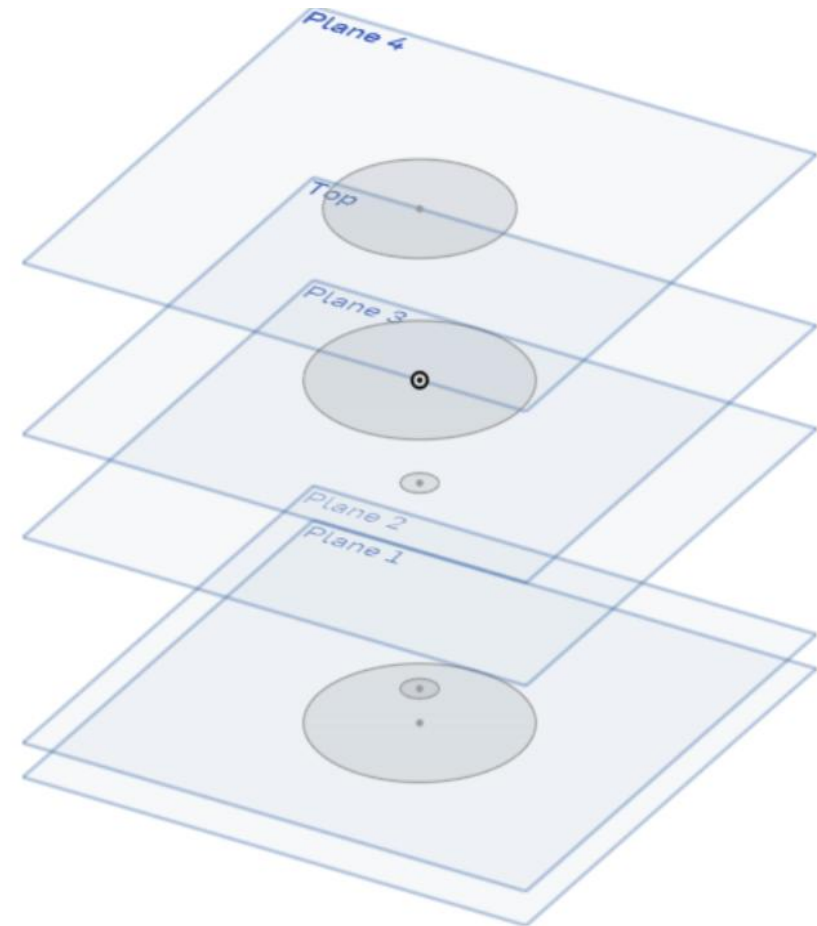
Note: To **hide** planes, place the clicker on top of it on the left, and click the eye next to it. For features, right click them and select suppress. To **unhide**, repeat the same steps.



# Wine Glass: Sketch

For this 3D model, we are only going to **sketch circles**. Draw a circle with diameter of

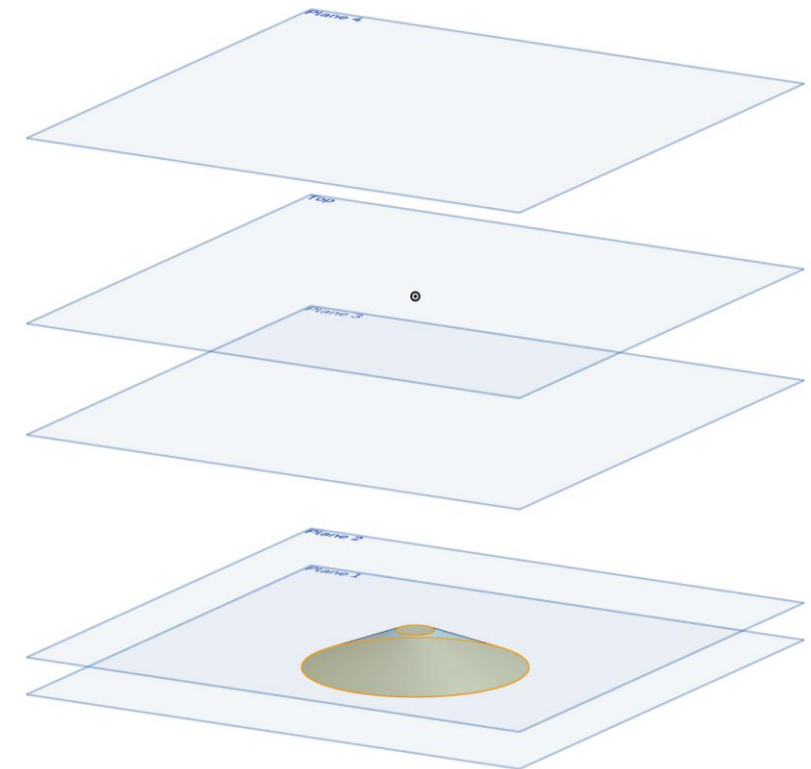
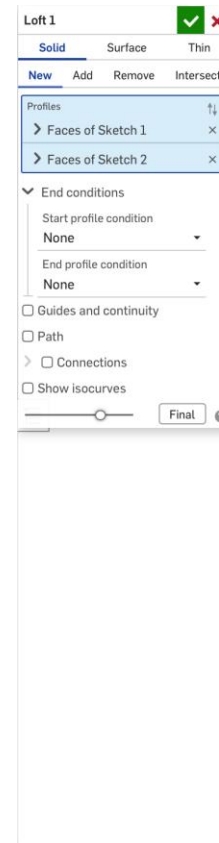
1. 6 cm for Plane 1
2. 1 cm for Planes 2
3. 1 cm for Plane 3
4. 6 cm for Top plane
5. 5 cm for Plane 4



# Wine Glass: Loft

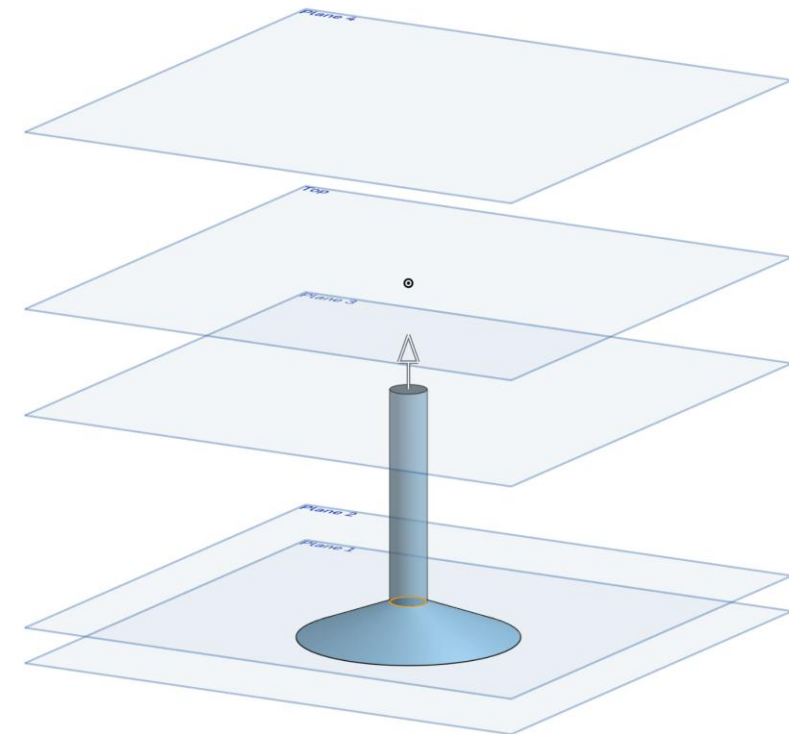
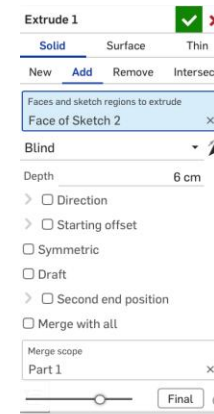


Select sketch 1 and 2, and then click on the **loft feature**. The loft feature will connect the two circles to create a 3D shape. Click the green check mark / Enter to finish.



# Wine Glass: Extrude

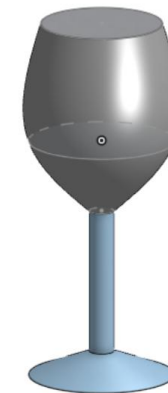
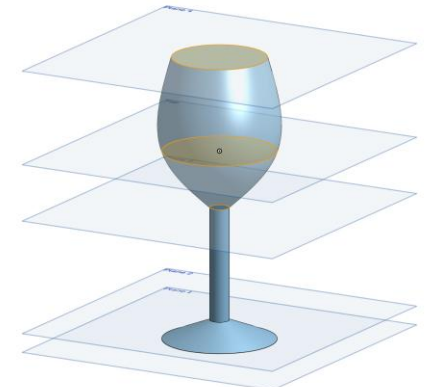
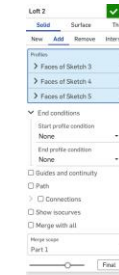
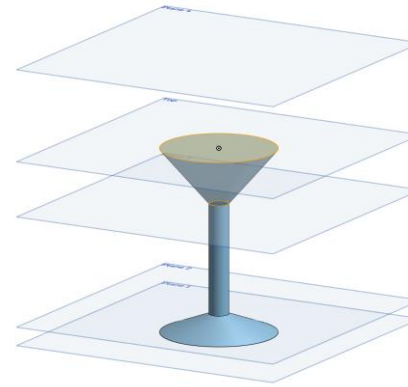
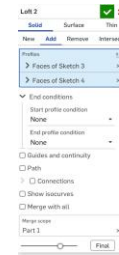
Click **extrude** and select the top (smaller) **face** of the **loft**. Change the **depth** to 6 cm so it reaches plane 3.



# Wine Glass: More Loft

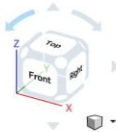
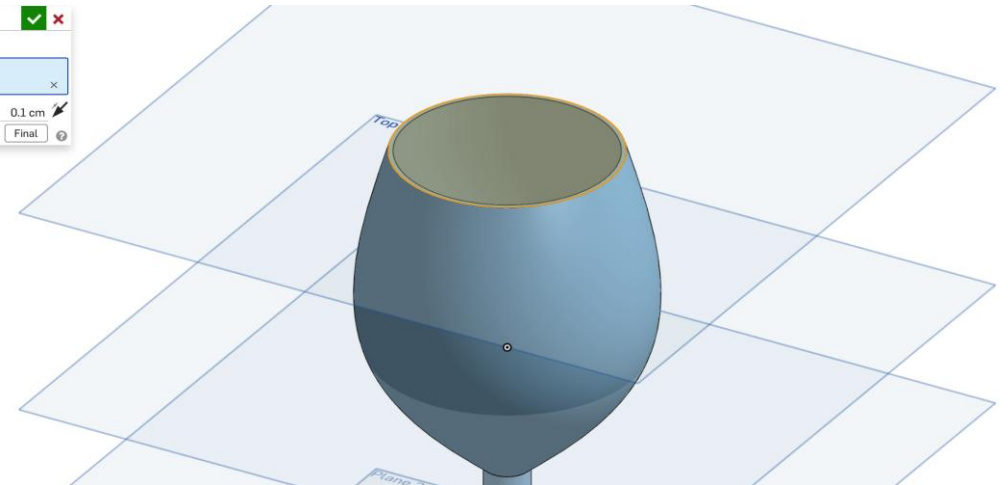
Click the **loft** feature, then select sketch 3 and 4 and hit the **enter** key. To edit the **loft**, you can **double click** it on the left side and add sketch 5.

Note: You should click edit to make the loft one part. If you originally add the 3 sketches to make the loft, it is going to do the loft as a separate part.



# Wine Glass: Shell

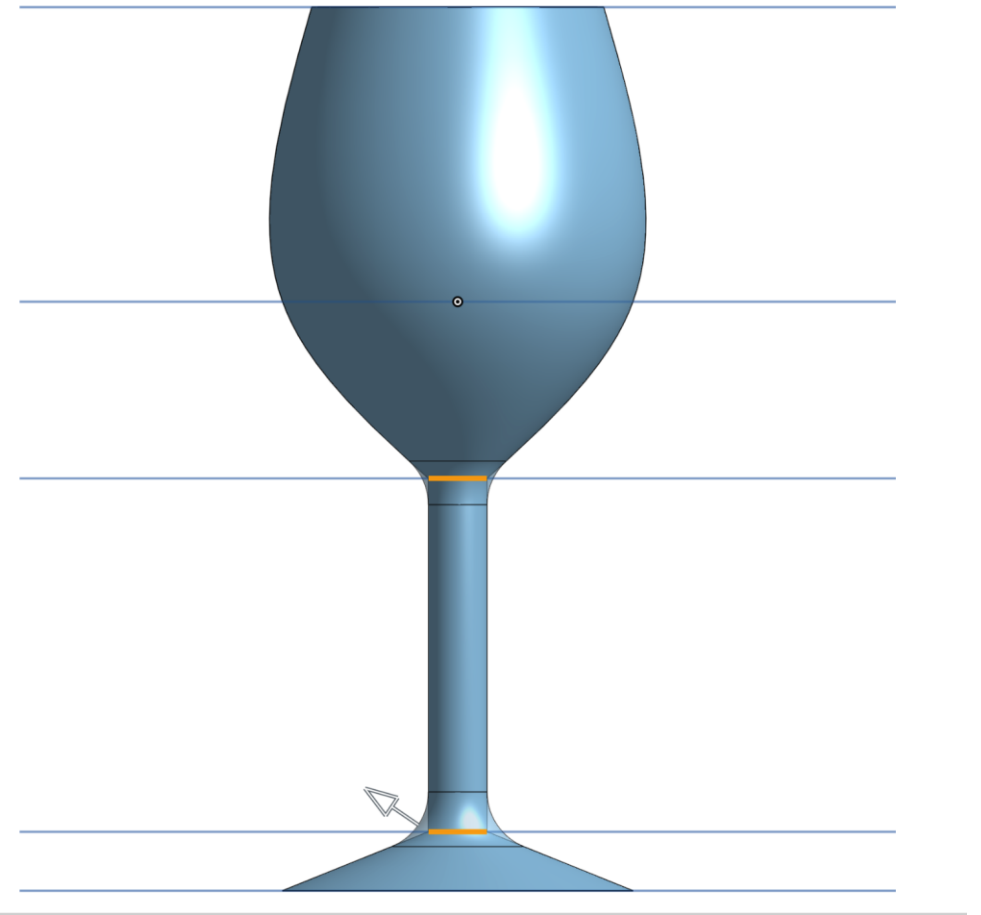
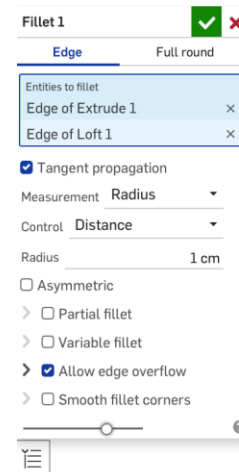
Click the **shell** feature, select the **top** of the wine glass, and set the **shell thickness** to 0.1 cm.





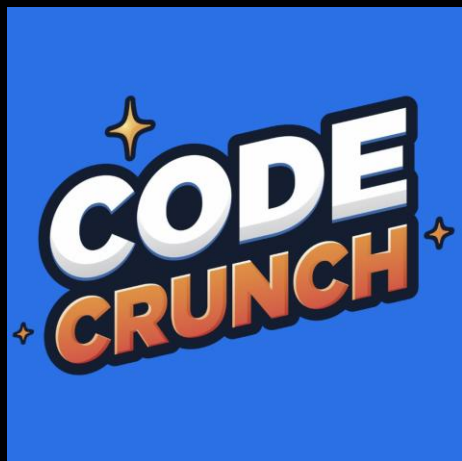
# Wine Glass: Fillet

Click the **fillet** feature, select the **bottom and top circles of the handle**, and set the **radius** to 1 cm.





# Q&A Session



# Attendance



Join us for the upcoming weekly sessions this Spring 2025!  
*(Jan 6 - Apr 4, 2025)*



**Monday:** C5 1PM  
**Tuesdays:** C7 1PM / C11 3PM  
**Wednesdays:** C5 12:30 PM / C9 1:30 PM (Varies)  
C1 2:30PM / C2 3:30PM / C13 5:30PM  
**Thursdays:** C6 1PM / C8 3PM  
**Friday:** C9 2PM (Varies) / C10 3PM/ C11 4PM



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